

Efficacy and Safety of Psilocybin-Assisted Therapy for Alcohol Use Disorder

A Systematic Review and Meta-Analysis

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Rationale

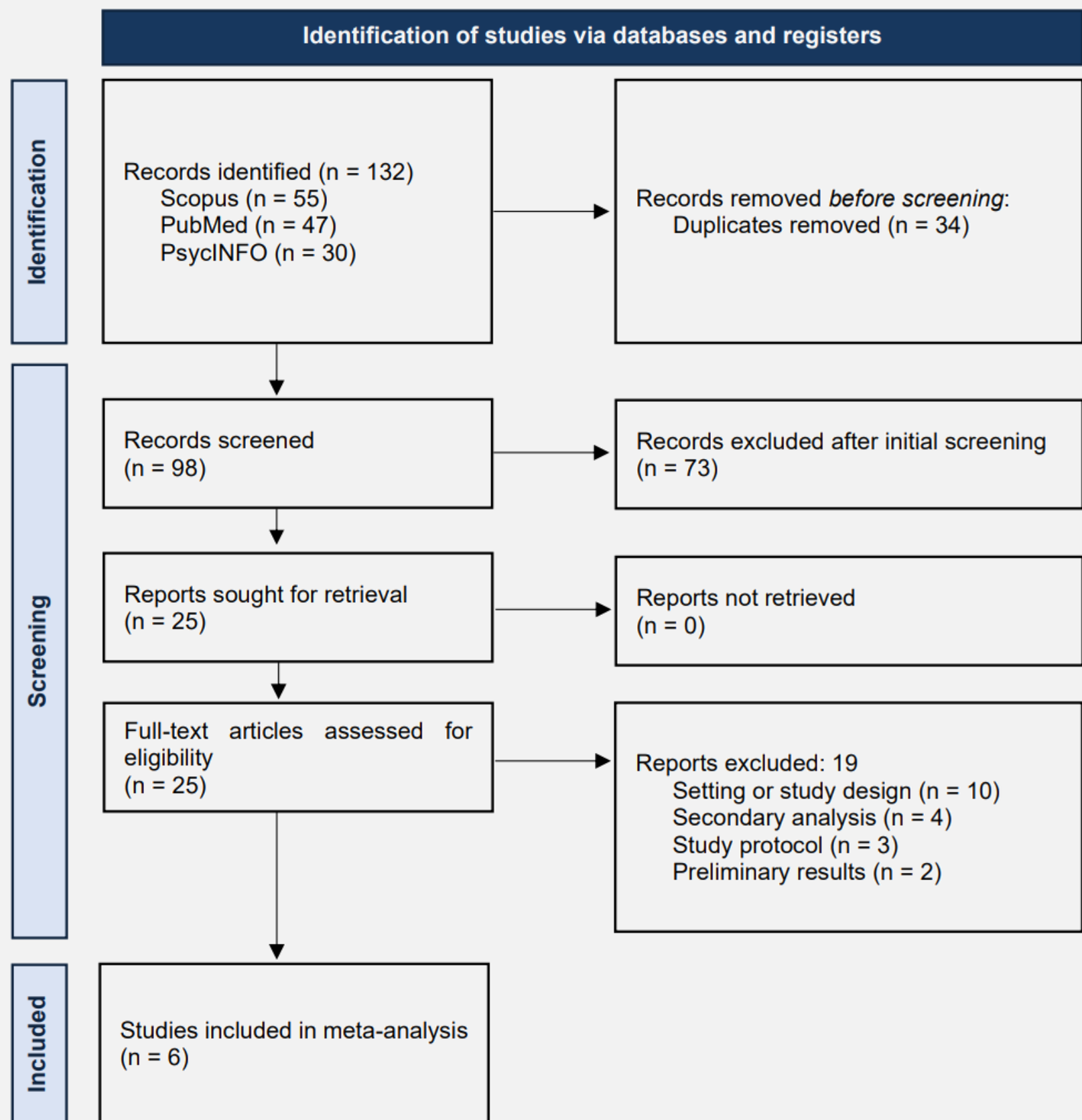
- Alcohol use disorder (AUD) is a major contributor to global morbidity and premature mortality with a lack of effective treatment options ^[1]
- Psilocybin-assisted therapy (PAT) has reemerged as a promising intervention for AUD, yet contemporary clinical evidence has not been systematically synthesized to validate its clinical utility and safety profile ^[2]

Aim

To pool and evaluate the efficacy and safety outcomes from contemporary clinical trials investigating PAT for AUD

Method

- Following PRISMA guidelines, we synthesized data from all contemporary clinical trials (01/2000 until 01/2026) where patients with AUD received PAT
- Primary outcomes included continuous drinking measures (drinks per day, percent drinking days, and percent heavy drinking days) and abstinence rates; secondary outcomes included adverse events.
- Outcomes were analyzed using standardized mean change (SMC), standardized mean differences (SMD), and odds ratios (OR).



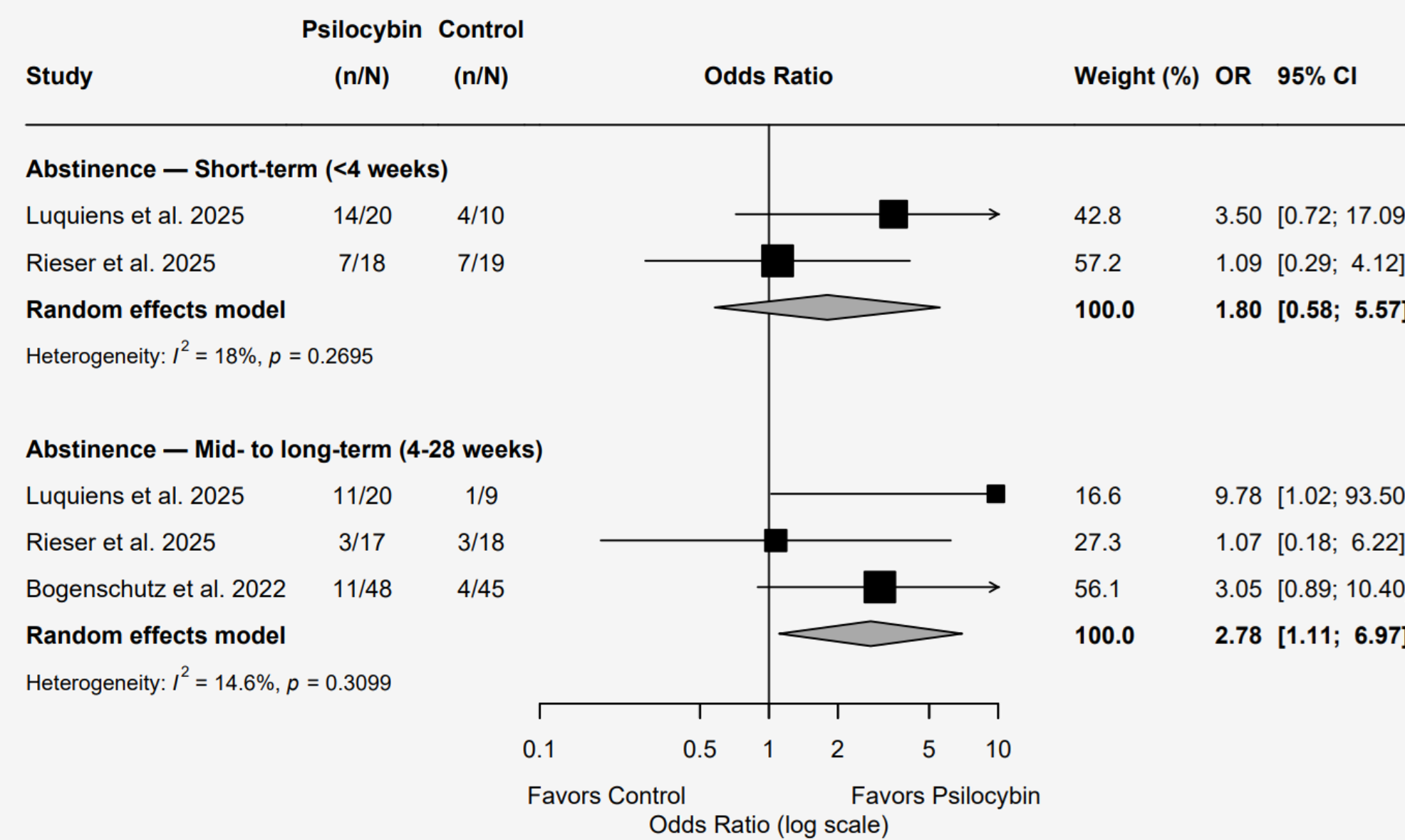
Results

- Six trials ($N = 202$) met inclusion criteria (3 RCTs) ^[3-8]
- PAT produced large, sustained pre-post reductions in alcohol consumption and demonstrated moderate short-term superiority over controls for reducing drinking

Analysis Type	Timepoint	Outcome Measure	Effect Size	95% CI
Within-Group (Pre vs. Post)	Short-Term (< 4 weeks)	Drinks per day	SMC = 1.27	0.90 to 1.63
		% Drinking days	SMC = 1.55	1.02 to 2.08
		% Heavy drinking days	SMC = 1.59	1.30 to 1.87
	Mid- to Long-Term (4–28 weeks)	Drinks per day	SMC = 1.13	0.78 to 1.48
		% Drinking days	SMC = 1.39	0.80 to 1.97
		% Heavy drinking days	SMC = 1.47	1.19 to 1.76
Between-Group (Psilocybin vs. Control)	Short-Term (< 4 weeks)	Drinks per day	SMD = 0.48	0.15 to 0.81
		% Drinking days	SMD = 0.39	0.02 to 0.77
		% Heavy drinking days	SMD = 0.37	0.04 to 0.69
	Mid- to Long-Term (4–28 weeks)	Drinks per day	SMD = 0.14	−0.35 to 0.63
		% Drinking days	SMD = 0.43	0.05 to 0.80
		% Heavy drinking days	SMD = 0.10	−0.46 to 0.67

SMC = standardized mean change, SMD = standardized mean difference

- Psilocybin treated patients showed significant long-term benefits for abstinence, with nearly three times higher odds of maintaining abstinence compared to controls



- PAT demonstrated a favorable safety profile, with no serious adverse events directly attributed to psilocybin

Study	Psilocybin treatment			Control treatment			Total		
	Patients	AEs	SAEs	Patients	AEs	SAEs	Patients	AE	SAE
Luquiens et al. (2025)	20	19	0	10	6	0	30	25	0
Rieser et al. (2025)	18	13	1	19	7	4	37	20	5
Jensen et al. (2025)	10	46	0	-	-	-	10	46	0
Heinzerling et al. (2023) ^a	20	63	2	- ^a	- ^a	- ^a	20	63	2
Bogenschutz et al. (2022)	49	119	0	46	85	3	95	204	3
Bogenschutz et al. (2015)	10	8	0	-	-	-	10	8	0

AE = adverse event, SAE = serious adverse event.

^a In Heinzerling et al. (2023), all 63 AEs were counted under psilocybin treatment as both randomized groups received psilocybin.

Conclusion

- PAT represents a promising, safe, and effective novel intervention for AUD, particularly for patients seeking total abstinence
- Our findings suggest that while psilocybin significantly reduces drinking intensity, its long-term comparative efficacy may depend heavily on the quality and intensity of the accompanying psychological therapy
- High response rates in control groups highlight the challenge of demonstrating added value of PAT in addiction trials
- Future research should prioritize large, rigorous trials with long-term follow-ups and aim to optimize the therapeutic framework of PAT for AUD

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